

GOPI JAGADHEESH MAHAMKALI

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PROFESSIONAL SUMMARY

Software Engineer Intern with hands-on backend and AI engineering experience building production RAG pipelines and document-automation systems at Qualizeal and 4SightAI. CS undergraduate (CGPA **8.99**) strong in Python, REST API design, and DS&A, seeking a Software Engineer Intern role focused on scalable backend systems.

TECHNICAL SKILLS

Languages: Python, C++, SQL.

Software Engineering: RESTful API Design, Backend Development, Microservices Architecture, OOP's, System Design.

Frameworks & Libraries: FastAPI, React, Vite, LangChain, LangGraph, scikit-learn, OpenCV.

Databases: PostgreSQL, MongoDB, MySQL, SQLite, Pinecone DB.

Tools: Git, GitHub, Docker, Swagger, Postman, Google Colab, RoboFlow, Kaggle, MLFlow, Jira, Azure DevOPs.

AI & ML: Machine Learning, Deep Learning, Computer Vision, Generative AI, LLM Integration, RAG, MCP.

WORK EXPERIENCE

Qualizeal

June 2026 - Present

Software Development Engineer - Intern (AI & Backend)

Hyderabad, India

- Delivered core backend modules for **ValidAite**, an enterprise AI-testing platform, implementing the RMTE evaluation framework (Risk, Metrics, Testing, Evidence) across **12 enterprise use cases** by designing modular test-orchestration **REST APIs in FastAPI**.
- Engineered a **RAG-based** backend pipeline that auto-generates prompts and ground-truth answers directly from uploaded documents, cutting manual QA-pair authoring time by **60%** through automated test-case creation in the ValidAite product.

4SightAI

October 2025 - March 2026

Software Engineer Intern (AI & Backend)

Vijayawada, India

- Led software design and development of **DOCS2DATA**, a core module of the AI4Andhra Police Pilot Project, reducing manual document processing time by **90%** by automating structured data extraction with unit-validated pipelines.
- Built document processing pipelines integrating **Azure Document Intelligence** with rule-based validation logic, improving extraction accuracy across diverse police department document formats.
- Engineered a custom-trained **YOLO-based** signature and stamp detection module, achieving **97% detection precision** on scanned police department documents.
- Built a multi-agent AI system combining **YOLO + Google Gemini** for real-time road safety analysis, reducing manual review effort by **90%** under project RTGS of Andhra Pradesh.

PROJECTS

Community Hero — Civic AI Co-Pilot — Python, FastAPI, MongoDB, React

- Built a production-ready AI civic collaboration platform using **React (Vite)**, **FastAPI**, **MongoDB**, and **SQLite**, featuring modular repository-based architecture, role-based task management, and scalable REST APIs.
- Engineered an LLM-powered multimodal inspection pipeline with **Google Gemini 1.5 Flash** for automated hazard classification, geospatial duplicate detection, and AI-based completion verification from before/after images.

Drone-Based Weapon Detection System — Team of 2 — Python, PyTorch, YOLOv8, FastAPI

- Developed a low-latency, two-stage inference pipeline for real-time computer vision, integrating person detection (**YOLOv8n**) with a localized classification module to identify target weapons within detected regions of interest.
- Optimized inference workloads through region-of-interest processing, reducing frame-processing latency and eliminating **40%** of background-induced false positives.

EDUCATION

Rajiv Gandhi University of Knowledge Technologies, Nuzvid

September 2023 - Present

Bachelor of Technology in Computer Science & Engineering (Expected May 2027)

CGPA: **8.99**

Rajiv Gandhi University of Knowledge Technologies, Nuzvid

December 2021 - August 2023

Pre-University Certificate (PUC)

CGPA: **9.85**

ACHIEVEMENTS

- Secured **All India Rank 3863** in GATE Data Science & AI (2026) assessing CS fundamentals and ML theory.
- Won **1st place** in **AI4Andhra Police Hackathon** for DOCS2DATA — GenAI-based document workflow automation.
- Solved **200+ problems** on **LeetCode**, covering core Data Structures & Algorithms.
- Finalist in HackRx 6.0 — GenAI hackathon conducted by **Bajaj Finserv**, competing among **1000+ teams** nationally.